

A COMPACT L -SPACE UNDER CH

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The continuum Hypothesis implies that there is a compact Hausdorff space which is hereditarily Lindelöf but not separable. The space is the support of a Borel probability measure for which the measure-0 subsets, the first-category subsets, and the separable subsets all coincide.

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L -space Borel measure

0. Introduction

An L -space is a T_3 space which is hereditarily Lindelöf (HL), but not hereditarily separable. See [4, 5 or 6] for a general discussion of such spaces. The purpose of this paper is to show that the Continuum Hypotheses (CH) implies that there is a compact L -space. One cannot produce a compact L -space from the usual axioms for set theory (ZFC) alone, since Juhász (see [4, p. 63]) showed that Martin's Axiom plus not CH implies that there are no such spaces.

A closed interval in a Suslin line is a well-known example of a compact L -space, but one cannot prove from CH that there is a Suslin line (Jensen: see [1]). Also, the space we produce here will have some interesting measure-theoretic properties which a Suslin line cannot have.

1. The space

Our plan is to modify the method of van Douwen, Tall, and Weiss [2] for producing L -spaces under CH. They observed that if X is T_2 , has no isolated points, and is a *Luzin space* (all nowhere dense sets countable), then X is HL. Now, such a space cannot be compact, so we weaken the notion of Luzin.

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Definition 1.1. An *almost Luzin space* is one in which every nowhere dense set is second countable in its relative topology.

Lemma 1.2. If X is T_2 , has no isolated points, and is almost Luzin, then X is HL.

Proof. First, observe that X has the countable chain condition (c.c.c.), since if $\{U_\alpha : \alpha < \omega_1\}$ were a disjoint family of non-empty open sets, we could pick $p_\alpha \in U_\alpha$, and obtain $\{p_\alpha : \alpha < \omega_1\}$, which would be nowhere dense and discrete.

Next, recall that X is HL iff whenever $\mathcal{U}$ is a family of open subsets of X , there is a countable $\mathcal{V} \subset \mathcal{U}$ with $\bigcup \mathcal{V} = \bigcup \mathcal{U}$. Now, if $\mathcal{U}$ is such a family, we may apply the c.c.c. to get a countable $\mathcal{V}_1 \subset \mathcal{U}$ with $\bigcup \mathcal{V}_1$ dense in $\bigcup \mathcal{U}$. But then $(\bigcup \mathcal{U}) \setminus (\bigcup \mathcal{V}_1)$ is nowhere dense and hence second countable, so there is a countable $\mathcal{V}_2 \subset \mathcal{U}$ with $(\bigcup \mathcal{U}) \setminus (\bigcup \mathcal{V}_1) \subset \bigcup \mathcal{V}_2$. If $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$, then $\bigcup \mathcal{V} = \bigcup \mathcal{U}$.

Any compact metric space is a trivial example of a compact almost Luzin space, but such a space is also separable. Our X will have the additional property that every separable subset of X is nowhere dense. One example of such an X is a closed interval in a Suslin line, but we shall construct our X just assuming CH. Our X will also be the support of a Borel measure μ and have the property that for all $Y \subset X$, Y is separable iff $\mu(Y) = 0$. Thus, X cannot come from a Suslin line, since any Borel measure on a Suslin line has a separable support.

We now review some conventions regarding measures. For simplicity, we consider only the case when X is compact T_2 and 0-dimensional. Then the *Baire* subsets of X are those in the least σ -algebra containing the clopen sets, while the *Borel* subsets of X are those in the least σ -algebra containing the closed sets. If X is HL, then every closed set is a G_δ , so these σ -algebras are the same, but it will be more convenient to deal with the Baire sets until we have proven that X is HL. A *Baire probability measure* on X is a (σ -additive) measure μ on the Baire sets with $\mu(X) = 1$. If $Y \subset X$, $\mu(Y) = 0$ means that $\mu(Z) = 0$ for some Baire Z with $Y \subseteq Z$. It is well known (see [3, Section 52]) that every such measure is regular; i.e., if $Z \subset X$ is a Baire set, then $\mu(Z)$ is the sup of the measures of the closed G_δ subsets of Z and is also the inf of the measures of the open F_σ supersets of Z .

The hypothesis of the next lemma lists the properties of the X we shall construct.

Lemma 1.3. Let X be compact T_2 and 0-dimensional, μ a Baire probability measure on X , and suppose that

- (i) for all $p \in X$, $\mu(\{p\}) = 0$,
- (ii) for all non-empty clopen $K \subset X$, $\mu(K) > 0$,
- (iii) for all nowhere dense $Y \subset X$, $\mu(Y) = 0$, and
- (iv) for all nowhere dense $Y \subset X$, Y is second countable.

Then X is an L-space. Furthermore, for all $Y \subset X$, the following are equivalent:

- (a) Y is nowhere dense in X .
- (b) $\mu(Y) = 0$.

- (c) Y is second countable.
- (d) Y is separable.
- (e) Y is first category in X .

Proof. X is almost Luzin by (iii). X has no isolated points by (i) and (ii), and is thus HL by Lemma 1.2. That X is not separable will follow from the fact that (d) $\Rightarrow$ (a).

To prove that (a)–(e) are equivalent, note first that (a) $\Rightarrow$ (b) is trivial. To prove (b) $\Rightarrow$ (a), assume $\mu(Y) = 0$. Let U_n ($n \in \omega$) be open F_σ sets with $Y \subset \bigcap_n U_n$ and $\mu(U_n) < 0$. By (iii), $\mu(\bar{U}_n \setminus U_n) = 0$, so if $F = \bigcap_n \bar{U}_n$, then $Y \subset F$ and $\mu(F) = 0$. By (ii), $F^0 = \emptyset$, so Y is nowhere dense.

We now have (a) $\Leftrightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d). To prove (d) $\Rightarrow$ (a), fix $D \subset Y$ with D dense in Y and D countable. By (i), $\mu(D) = 0$, so by (b) $\Rightarrow$ (a), D is nowhere dense in X . Hence, Y is nowhere dense. Finally, (a) $\Rightarrow$ (e) is trivial and (e) $\Rightarrow$ (b) follows from (a) $\Rightarrow$ (b).

Theorem 1.4. CH implies that there is a compact L-space.

Proof. We shall construct X and μ satisfying (i)–(iv) of Lemma 1.3. In doing so, we inductively construct spaces X_α for $\omega \leq \alpha \leq \omega_1$ and Baire probability measures μ_α on X_α ; then μ will be μ_{ω_1} and X will be X_{ω_1} . Each X_α will be a closed subset of 2^α . Thus, for $\alpha < \omega_1$, all Borel subsets of 2^α will be Baire. If $\alpha \leq \beta$, let π_α^β be the natural projection from 2^β onto 2^α ; thus, $\pi_\alpha^\beta(f) = f \upharpoonright \alpha$. We shall demand of our construction that:

(0) If $\alpha \leq \beta$, then $X_\alpha = \pi_\alpha^\beta(X_\beta)$ and $\mu_\alpha(Y) = \mu_\beta((\pi_\alpha^\beta)^{-1}Y)$ for any Baire subset Y of X_α .

Thus, we are demanding that μ_α is the measure induced from μ_β via the map π_α^β .

We also demand that each X_α , μ_α satisfy the ‘trivial’ conditions (i) and (ii) of Lemma 1.3; namely:

- (1) For each α , $\mu_\alpha(\{p\}) = 0$ for all $p \in X_\alpha$, and
- (2) For each α , $\mu_\alpha(K) > 0$ for all non-empty clopen $K \subset X_\alpha$.

Observe that the construction is determined at limit ordinals $\gamma > \omega$. Thus, by compactness and (0), we must have

$$X_\gamma = \{f \in 2^\gamma : \forall \alpha < \gamma (f \upharpoonright \alpha \in X_\alpha)\}.$$

Also by compactness, if K is a clopen subset of X_γ , then for some $\alpha < \gamma$ and some clopen $L \subset X_\alpha$, $K = (\pi_\alpha^\gamma)^{-1}(L)$; we define $\mu(K)$ to be $\mu_\alpha(L)$; by (0), $\mu(K)$ does not depend upon the α we use. μ is thus defined on the clopen subsets of X_γ and is easily seen to be finitely additive, and hence, by compactness, countably additive, so μ extends uniquely to a Baire probability measure, μ_γ on X_γ . Conditions (1) and (2) are clearly preserved. We remark that X_γ may be viewed as an inverse limit of $\{X_\alpha : \alpha < \gamma\}$. Also, our construction of μ_γ was essentially the proof of a special case of the Kolmogorov Extension Theorem.

Next, observe that there is no problem starting the construction. We set $X_\omega = 2^\omega$, and let μ_ω be the usual product measure on 2^ω (where in $2 = \{0, 1\}$, $\{0\}$ and $\{1\}$ are given probability $\frac{1}{2}$).

Thus, the "meat" of our construction lies in the successor steps. We shall attain condition (iii) of Lemma 1.3 by ensuring that closed sets of positive measure eventually obtain an interior, and we shall attain (iv) by ensuring that closed nowhere dense sets only get split countably often. All this requires CH to enumerate, in type ω_1 , all relevant sets.

Applying CH, fix, at the outset, an enumeration, $\{F_\xi^\alpha : \xi < \omega_1\}$, of all closed subsets of 2^ω when $\omega \leq \alpha < \omega_1$. Also, fix a map g from ω_1 onto $\omega_1 \times \omega_1$ such that $g(\beta) = \langle \alpha, \xi \rangle$ implies $\alpha \leq \beta$.

Condition (iii) of Lemma 1.3 will be handled by demanding:

(3) For each $\beta < \omega_1$: If $g(\beta) = \langle \alpha, \xi \rangle$, $F_\xi^\alpha \subset X_\alpha$, and $\mu_\alpha(F_\xi^\alpha) > 0$, then there is a closed $C \subset (\pi_\alpha^\beta)^{-1}(F_\xi^\alpha)$ such that $\mu_\beta(C) > 0$ and

$$X_{\beta+1} = X_\beta \times \{0\} \cup C \times \{1\}.$$

Here, we are making the obvious identification of $2^{\beta+1}$ with $2^\beta \times 2$. Condition (iv) of Lemma 1.3 will be handled by demanding:

(4) For each $\beta < \omega_1$, and all $\alpha, \xi \leq \beta$: If $F_\xi^\alpha \subset X_\alpha$ and $\mu_\alpha(F_\xi^\alpha) = 0$, then

$$X_{\beta+1} \cap ((\pi_\alpha^\beta)^{-1}(F_\xi^\alpha) \times \{1\}) = \emptyset.$$

Let us check that (3) and (4) are indeed possible. Fix $\beta < \omega_1$, and let $\langle \alpha, \xi \rangle = g(\beta)$. If it is not the case that both $F_\xi^\alpha \subset X_\alpha$ and $\mu_\alpha(F_\xi^\alpha) > 0$, then (3) is vacuous and we satisfy (4) by taking $X_{\beta+1} = X_\beta \times \{0\}$. Now assume $F_\xi^\alpha \subset X_\alpha$ and $\mu_\alpha(F_\xi^\alpha) > 0$, and let

$$D = (\pi_\alpha^\beta)^{-1}(F_\xi^\alpha) \setminus \bigcup \{(\pi_\delta^\beta)^{-1}(F_\eta^\delta) : \delta, \eta \leq \beta \text{ and } F_\eta^\delta \subset X_\delta \text{ and } \mu_\delta(F_\eta^\delta) = 0\}.$$

D is a Baire set of positive μ_β measure, so let $E \subset D$ be closed with $\mu_\beta(E) > 0$. Let

$$C = E \setminus \bigcup \{K \subset X_\beta : K \text{ is clopen and } \mu_\beta(E \cap K) = 0\}.$$

Then $\mu_\beta(C) = \mu_\beta(E) > 0$ and

$$\forall K \subset X_\beta [(K \text{ clopen and } \mu_\beta(C \cap K) = 0) \Rightarrow C \cap K = \emptyset]. \quad (*)$$

Let $X_{\beta+1} = X_\beta \times \{0\} \cup C \times \{1\}$, and define $\mu_{\beta+1}$ by splitting C in half; thus, if $A \subset C$,

$$\mu_{\beta+1}(A \times \{0\}) = \mu_{\beta+1}(A \times \{1\}) = \frac{1}{2}\mu_\beta(A),$$

and if $A \subset X_\beta \setminus C$, $\mu_{\beta+1}(A \times \{0\}) = \mu_\beta(A)$. It is easily checked that (0)–(4) are preserved; (*) is used to show that non-empty clopen subsets of $C \times \{1\}$ have positive measure.

Now, let $X = X_{\omega_1}$ and $\mu = \mu_{\omega_1}$, and we verify that (i)–(iv) of Lemma 1.3 do indeed hold. (i) and (ii) are immediate. For (iii), assume $Y \subset X$ is nowhere dense. By (ii), X has the c.c.c., so there is a nowhere dense closed G_δ , $G \subset X$, with $Y \subset G$. Since G is the intersection of countably many clopen sets, there is an $\alpha < \omega_1$ such that

$(\pi_\alpha^{\omega_1})^{-1}(F)$ for some closed $F \subset X_\alpha$. Say $F = F_\xi^\alpha$, and fix $\beta \geq \alpha$ with $g(\beta) = \langle \alpha, \xi \rangle$. If $\mu_\alpha(F) > 0$ and C is as in (3), then $(\pi_{\beta+1}^{\omega_1})^{-1}(C \times \{1\})$ would be a clopen subset of G ; thus $\mu_\alpha(F) = 0$. Hence, $\mu(G) = 0$ so $\mu(Y) = 0$.

For (iv), let Y, G, F, α , and ξ be as above; then, as we have seen, $\mu_\alpha(F) = 0$. Let $\delta = \max(\alpha, \xi)$. If $\beta \geq \delta$, let $F_\beta = (\pi_\alpha^\beta)^{-1}(F)$. By (4) of our construction,

$$(\pi_{\beta+1}^{\beta+1} \upharpoonright F_{\beta+1}) : F_{\beta+1} \rightarrow F_\beta$$

is a one-to-one map whenever $\delta \leq \beta < \omega_1$, so, by induction on β ,

$$(\pi_\delta^\beta \upharpoonright F_\beta) : F_\beta \rightarrow F_\delta$$

is one-to-one, and hence a homeomorphism, whenever $\delta \leq \beta \leq \omega_1$. In particular, $G = F_{\omega_1}$ is homeomorphic to F_δ , so G , and hence Y , is second countable.

In conclusion, we comment on the measure algebra, $\langle \mathcal{B}, \mu \rangle$, of $\langle X, \mu \rangle$. In our construction, sets of positive measure got split cofinally often, so $\langle \mathcal{B}, \mu \rangle$ is nowhere separable. Thus, by Maharam's Theorem it is isomorphic to the usual measure algebra of 2^{ω_1} . However, we may modify the construction to make $\langle \mathcal{B}, \mu \rangle$ separable. To do this, we define $\mu_{\beta+1}$ in the case that the hypotheses to (3) hold so that for $A \subset C$,

$$\mu_{\beta+1}(A \times \{0\}) = 0 \quad \text{and} \quad \mu_{\beta+1}(A \times \{1\}) = \mu_E(A);$$

this will require that E (and hence C) is taken to be nowhere dense in X_β so that (2) is preserved. Then $\pi_\omega^{\omega_1}$ defines an isomorphism of $\langle \mathcal{B}, \mu \rangle$ onto the measure algebra of 2^ω .

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